

Linear Algebra Recitation

Week 5

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Matrix Inversion

$\in \mathbb{R}^{3 \times 3}$

$$AB=I \quad B=[u|v|w], \quad I=[e_1|e_2|e_3]$$

$$BA=I \quad Ax=b \quad x=A^{-1}b$$

- Let A be invertible. Then finding the inverse A^{-1} is solving three system of equations:

$$B=[u|v|w]=[A^{-1}e_1|A^{-1}e_2|A^{-1}e_3]=A^{-1}[e_1|e_2|e_3]$$

$$[A|e_1] \quad [A|e_2] \quad [A|e_3]$$

$$u=A^{-1}e_1 \quad v=A^{-1}e_2 \quad w=A^{-1}e_3$$

$$Au=e_1, \quad Av=e_2, \quad Aw=e_3,$$

$$B=A^{-1}I=A^{-1}$$

where e_k is the k^{th} standard unit vector. If we can find such vectors u, v, w , then

$$[A|e_1|e_2|e_3]=[A|I] \sim [RREF|u|v|w] \quad A[u|v|w]=[u|v|w]A=I$$

and we call $A^{-1}=[u|v|w]$.

- It is clear that u, v, w can be obtained from Gauss elimination.
- Note that the inverse of a matrix is only relevant to a square matrix.

Matrix Inversion

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

- Basic example: Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$. Find A^{-1} using Gauss

elimination.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 2 & 1 & 3 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 1 & 0 \\ 0 & -3 & 1 & -2 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & -1 & 0 \\ 0 & 0 & -2 & 1 & -3 & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1/2 & 3/2 & -1/2 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 3/2 & -3/2 & 1/2 \\ 0 & 1 & 0 & 3/2 & -5/2 & 1/2 \\ 0 & 0 & 1 & -1/2 & 3/2 & -1/2 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1/2 & 7/2 & -1/2 \\ 0 & 1 & 0 & 3/2 & -5/2 & 1/2 \\ 0 & 0 & 1 & -1/2 & 3/2 & -1/2 \end{array} \right]$$

REF = I U V W = B = A⁻¹

check if this holds!

$$AB = I$$

$$B = A^{-1}$$

The four fundamental subspaces $A: \mathbb{R}^p \rightarrow \mathbb{R}^2$, $\begin{bmatrix} p \\ A \end{bmatrix} \begin{bmatrix} 2 \\ A^T \end{bmatrix}$

- Recall the definition of orthogonal complement from last week.
- (Important) review the properties of orthogonal complements.

$$\star \text{col}(A^T)^\perp = \mathcal{N}(A)$$

$$\star \text{col}(A)^\perp = \mathcal{N}(A^T)$$

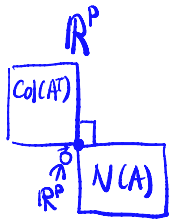
$$v \in \text{Col}(A^T), u \in \mathcal{N}(A)$$

$$v \in \mathbb{R}^p, u \in \mathbb{R}^p$$

$$v \cdot u = 0$$

$$\text{Col}(A^T) \cap \mathcal{N}(A) = \{\vec{0}\}$$

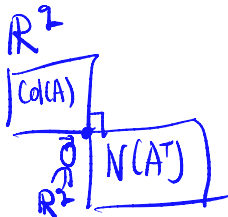
$$\underline{\dim \text{Col}(A^T) + \dim \mathcal{N}(A) = \dim \mathbb{R}^p}$$



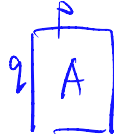
$$w \in \text{Col}(A), z \in \mathcal{N}(A^T)$$

$$w \in \mathbb{R}^2, z \in \mathbb{R}^2$$

$$\underline{\dim \text{Col}(A) + \dim \mathcal{N}(A^T) = \dim \mathbb{R}^2}$$



Review: rank-nullity theorem (a.k.a. the FTLA)



- rank-nullity theorem:

$$\dim \text{Col}(A^T) + \dim N(A) = \dim \mathbb{R}^p = p$$

$\text{Rank}(A) = \text{Rank}(A^T)$

$$\dim \text{Col}(A) + \dim N(A^T) = \dim \mathbb{R}^q = q$$

Codomain

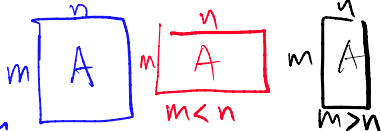
- recall the idea of proof:

the two spaces are orthogonal complements of each other +
dimension theorem of orthogonal complements.

Injective, Surjective, Bijective

Let $A \in \mathbb{R}^{m \times n}$

$$\exists x \in \mathbb{R}^n : Ax = y = x_1 a_1 + \dots + x_n a_n$$



- Rank-Nullity theorem is important (as always). If you cannot prove the " $\iff$ "s in each statement, take some time to think about it.

A is injective $\iff$ one to one $\iff \mathcal{N}(A)$ trivial $\iff \dim(\text{col}(A^T)) = n$ (full column rank)

$$\begin{aligned} Ax_1 &= b \\ Ax_2 &= b \end{aligned} \implies A(x_1 - x_2) = 0 \implies x_1 - x_2 = 0 \implies x_1 = x_2$$

A^T is injective $\iff \mathcal{N}(A^T)$ trivial $\iff \dim(\text{col}(A)) = m$

A is surjective $\iff \dim(\text{col}(A^T)) = m$ (full row rank)

A^T is surjective $\iff \dim(\text{col}(A)) = n$

Could you relate the injectivity/surjectivity of A and A^T ?

$m < n$, $m > n$, $m = n$. What happens in each case?

A cannot be injective

Surjective when $\text{rank}(A) = m$

A cannot be surjective

injective when $\text{rank}(A) = n$

injective

surjective

$\Rightarrow A$ is bijective

A bijective

$\iff$ injective + surjective

$\iff A$ invertible

System of linear equations: revisited $A(x_1 + x_n) = Ax_p + 0$

$$X = x_p + \underline{N(A)}$$

- c.f. last week's slides.

- In terms of injectivity/surjectivity/bijectivity,

- $Ax = b$ has a unique solution $\iff \begin{cases} N(A) \text{ is trivial} \iff A \text{ injective} \\ b \in \text{Col}(A) \end{cases}$

- $Ax = b$ has infinitely many solutions $\iff \begin{cases} N(A) \text{ is not trivial} \\ b \in \text{Col}(A) \end{cases} \iff A \text{ not injective}$

- $Ax = b$ has no solutions $\iff b \notin \text{Col}(A)$

$\implies A$ is not surjective
not iff!